



Noble gas isotopic constraints on the origin and evolution of the Jinchuan Ni–Cu–(PGE) sulfide ore-bearing ultramafic intrusion, Western China

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ABSTRACT

The Jinchuan Ni–Cu–PGE sulfide deposit in Western China is the largest single magmatic sulfide deposit hosted by a small ultramafic intrusion in the world. The abundances and isotopic compositions of He, Ne and Ar in 23 silicate and sulfide mineral separates from peridotites, pyroxenites and sulfide ores have been determined by stepwise heating to investigate the formation and evolution of the intrusion and associated sulfide ores. Intriguingly, the volatiles released from elevated temperatures (400–1300 °C) are all characterized by low $^3\text{He}/^4\text{He}$ ($< 1 \text{ Ra}$) that are close to crustal value, varying $^3\text{He}/^4\text{He}$, $^{21}\text{Ne}/^{22}\text{Ne}$, $^{20}\text{Ne}/^{22}\text{Ne}$ and $^{40}\text{Ar}/^{36}\text{Ar}$ ratios suggest a mixture of crust, mantle and air components. Numerical modeling reveals that the apparent “crustal signatures” of the noble gases released from the Jinchuan samples are due to significant radiogenic production from U and Th decay after the crystallization from magma derived from a subcontinental lithospheric mantle at ~830 Ma. In contrast, the noble gases released at low temperatures (200–400 °C) are characterized by abundant atmosphere-origin components, suggesting that they are mainly derived from secondary fluid inclusions. The noble gases in the 400–1300 °C fractions collectively exhibit a negative correlation between $^3\text{He}/^4\text{He}$ and ^4He , and a positive correlation between $^{40}\text{Ar}/^{36}\text{Ar}$ and ^{40}Ar , which are consistent with significant radiogenic in-growths. The sulfide separates have slightly higher $^3\text{He}/^4\text{He}$ ratios than coexisting silicates in the intrusion, indicating more abundant radiogenic components in the latter. Mass balance calculations of He and Ar isotopic mixing models indicate that about 20% crustal fluid and 63% air saturated fluid (ASF) were added into ore-forming magma. Our data are consistent with a recent suggestion by some researchers that significant amounts of crustal fluids were added to the parental magma of the Jinchuan intrusion.

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1. Introduction

The Jinchuan Ni–Cu–(PGE) sulfide deposit in Western China is the third largest magmatic sulfide deposit in the world. It is hosted by a small ultramafic intrusion. The Jinchuan deposit is remarkable and enigmatic to many exploration geologists and researchers due to its unusual geologic characteristics, including the small size of the host intrusion (6500 m long, 1100 m deep with an average width of 400 m), the high bulk sulfide-to-silicate ratio, contemporaneity with Rodinia break-up and related mantle plume activity, and controversial tectonic setting (Li et al., 2005; Zhang et al., 2010, 2011; Li and Ripley, 2011).

The accumulation of huge quantity of Ni- and Cu-rich sulfide liquid is a key to the formation of super-large Ni–Cu sulfide deposits in magma plumbing systems (e.g., Naldrett, 1999; Li and Ripley, 2005). The generation of large amounts of Ni–Cu sulfide liquids requires large amount of mantle-derived, fertile basaltic magma (Naldrett, 1999). Sulfur saturation in mantle-derived magma at the crustal levels commonly requires the addition of crustal sulfur or assimilation of country rocks (Li and

Ripley, 2005). A vast amount of Ni–Cu sulfide (5.45 million tons of metallic Ni) is present in the Jinchuan deposit. The host intrusion is a small ultramafic body surrounded by sulfur-poor country rocks (Tang et al., 2009). Sulfur isotopic data provide ambiguous evidence for the role of crustal sulfur in the genesis of Jinchuan ore deposit (e.g., Ripley et al., 2005). Recently, Lehmann et al. (2007) suggested that sulfide saturation in the Jinchuan magma was induced by addition of CO_2 -rich fluids from the country rock marbles during contact metamorphism.

Elemental and isotopic compositions of noble gases are excellent tracers of volatile sources, history and physical processes because of their rarity, chemical inertness and distinct isotopic compositions in different reservoirs in the Earth (e.g., Allegre et al., 1987; Norman and Musgrave, 1994; Burnard et al., 1998; Farley and Neroda, 1998; Moreira and Sarda, 2000; Ozima and Igarashi, 2000; Ozima and Podosek, 2002). Noble gas isotopes by a conventional heat-melting gas extraction hypothesized a mixture of the noble gases in different residence sites and sources (Hu et al., 2010; Hou et al., 2011). The accumulation of *in situ* radiogenic, secondary and cosmogenic components such as ^4He , ^{21}Ne and ^{40}Ar may erase the primary magmatic signatures if the intrusions are too old. Nonetheless, noble gas isotopic compositions have been successfully utilized to investigate the

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origin of volatiles in olivine and pyroxene from Archean komatiites (e.g., Richard et al., 1996; Matsumoto et al., 2002) and Permian continental flood basalts (Basu et al., 1993, 1995).

The crushing gas extraction is thought to be quite effective in preferentially releasing the noble gases trapped in inclusions, as radiogenic isotopes remain close to their production sites within crystal lattices. However, crushing can only crack and release the volatiles of large fluid inclusions and hence gives a small yield of volatiles (Landis and Rye, 2005; Kim et al., 2008). This method cannot extract volatiles from melt inclusions. In addition, intensive crushing may release radiogenic and cosmogenic components as well (Scarsi, 2000; Matsumoto et al., 2002; Yokochi et al., 2005). Stepwise heating can be more effective in separating volatiles of different occurrences (such as fluid versus melt inclusions) in a given sample (e.g., Macpherson et al., 1999; Villa, 2001; Kendrick et al., 2006; Zhang et al., 2007, 2009).

In this study, we used stepwise heating to extract volatiles from mineral separates from the Jinchuan sulfide ore-bearing intrusion. The noble gas abundances and isotopes were then determined by on-line mass spectrometry. The data are used to evaluate the source characteristics of the parental magma for the Jinchuan intrusion as well as the role of crustal fluids in ore formation at Jinchuan. This is the successful example of using noble gas isotopes to study the origin and evolution of a magmatic sulfide deposit worldwide.

2. Geological setting

2.1. Regional geology

The Jinchuan sulfide ore-bearing ultramafic intrusion is located in the northwest-striking, uplifted Longshoushan terrane in the southwestern portion of the North China Craton (Li and Ripley, 2011). It intruded the lower Proterozoic Baijiazui formation which is composed of granitic migmatites, gneisses and marbles, overlain by Late Mesoproterozoic sedimentary rocks. The overall strike of the intrusion is northwest. Its eastern and western parts are covered by Quaternary alluvium deposits (e.g., Tang and Li, 1995; Li et al., 2004; Lehmann et al., 2007) (Fig. 1a). The Jinchuan intrusion is divided into western, central, eastern and far-eastern segments (i.e., Mine III, I, II and IV, respectively) by a series of north-trending strike-slip (E–W to NE–SW) faults (Fig. 1b). Regional metamorphism of greenschist grade related to the collision event between the Qilian block and the North China Craton at ~450 Ma (Song et al., 2006b) has affected the southern part of the Longshoushan terrane where the Jinchuan intrusion is located.

2.2. Geology of the Jinchuan intrusion

The Jinchuan intrusion is composed almost entirely of olivine-rich ultramafic rocks, such as (in descending order of abundance) lherzolite, dunite, wehrlite and harzburgite, plus troctolite. Lherzolite and dunite commonly occur close to each other. In the middle segment of the intrusion dunite occurs in the center, surrounded by lherzolite. Harzburgite or wehrlite occurs in the margins. Minor troctolite and rare gabbroic rocks occur in the marginal in a few places. The contacts between different types of rocks are mostly gradational. The thermally annealed and chemically etched zircon and baddeleyite ID-TIMS U–Pb dating indicates that the intrusion is crystallized at 831.8 ± 0.6 Ma (Zhang et al., 2010).

2.3. Sulfide mineralization

The Jinchuan deposit contains more than 100 sulfide ore bodies. The largest 3 orebodies are orebodies 24, 1 and 2. They occur in the western (Mine I), central and eastern segments (Mine II), respectively. The largest sulfide orebody (Orebody 1) occurs in the central part of the intrusion ~300 m below the surface. The total amount of sulfide ores in the Jinchuan deposit is more than 500 million tons grading 1.1 wt.% Ni

and 0.7 wt.% Cu (Tang et al., 2009). The sulfide ores are mainly present in the form of disseminated and net-textured ores. Massive ores (>95% bulk sulfide) are rare.

In Orebody 2 the sulfide ores are concentrated in the lower part of the intrusion (Fig. 1c). Orebody 1 is characterized by a core of net-textured sulfides surrounded by disseminated sulfides and then by sulfide-poor lherzolite. Coexisting silicates in the sulfide ores are dominated by olivine. Hence the host rocks are called dunite. The sulfide assemblages are composed of pentlandite, chalcopyrite and pyrrhotite.

Both silicate (except phlogopite) and sulfide assemblages in the Jinchuan intrusion have experienced varying degrees of post-magmatic hydrothermal alteration (Ripley et al., 2005). Olivine is partially altered to serpentine plus secondary magnetite. Overall, olivine in dunite and sulfide-rich samples is more severely altered than olivine in lherzolite and sulfide-poor samples. Alteration is negligible for some types of rocks in some places, e.g., magmatic minerals are variably well preserved throughout the intrusion. Samples with fresh olivine crystals are present in a few places in the intrusion.

3. Sampling and analytical methods

3.1. Sample description

Samples with fresh olivine and pyroxene were collected mainly from outcrops and drill cores (ZK66, ZK71, ZK73, ZK80, ZK87 and ZK89) or from underground adits of orebodies 24, 1 and 2. The sample locations are shown in Fig. 1c. More detailed information of sample locations, latitude and longitude are given in dataset online (Table S1). The underground samples are most important because they are least or not affected by cosmic rays which may penetrate up to 1.5 m from the surface (e.g., Lal, 1987; Niedermann, 2002).

The sample types are dunite, lherzolite and pyroxenite and sulfide ores. More than 100 samples were selected for polished thin sections. The polished thin sections were carefully examined using transmitted light microscope to select the least altered samples for mineral separation.

Dunite is composed dominantly of cumulus olivine, with minor amounts of plagioclase, pyroxene (Pyx) and sulfides in the interstitial spaces. The most common olivine texture is a network of serpentine plus secondary magnetite in an olivine crystal. Lherzolite and olivine pyroxenite consist of olivine and pyroxene in variable proportions, with minor amounts of plagioclase, amphibole and phlogopite. In addition to distinct crystals, olivine is also present as relatively small inclusions within large clinopyroxene, orthopyroxene and plagioclase crystals (Fig. 2).

The relatively fresh samples were chosen in using thin sections under microscopic observation. The studied selected samples of ultramafic rocks and sulfide ores are neither least altered, nor deformed or metamorphosed. The rock samples are sulfide-poor lherzolite and pyroxenite, which are olivine, orthopyroxene and plagioclase cumulates. Some of the samples contain minor disseminated pyrrhotite, pentlandite and chalcopyrite (e.g., Fig. 2). The sulfide ore samples include disseminated, net-textured and massive ores.

Selected relatively fresh samples were cut and crushed to pieces 0.2–0.3 mm in diameter using a jaw crusher. Olivine (Olv), orthopyroxene (Opx), clinopyroxene (Cpx) and sulfide (Sfd) minerals were hand-picked under a binocular microscope. A total of 23 mineral separates from lherzolites, pyroxenite and net-textured sulfide ores (see Table S1) were used for noble gas isotopic analysis. The selected samples were immersed in 0.3 mol/L HCl for 24 h to remove possible carbonates and soluble alteration minerals that may occur in micro-fractures and cleavages. The samples were then washed using distilled water until the water reached a pH value around 7, and then cleaned ultrasonically with CH_2Cl_2 and acetone sequentially to remove possible organic contaminants. The cleaned samples were finally heated at 110 °C.

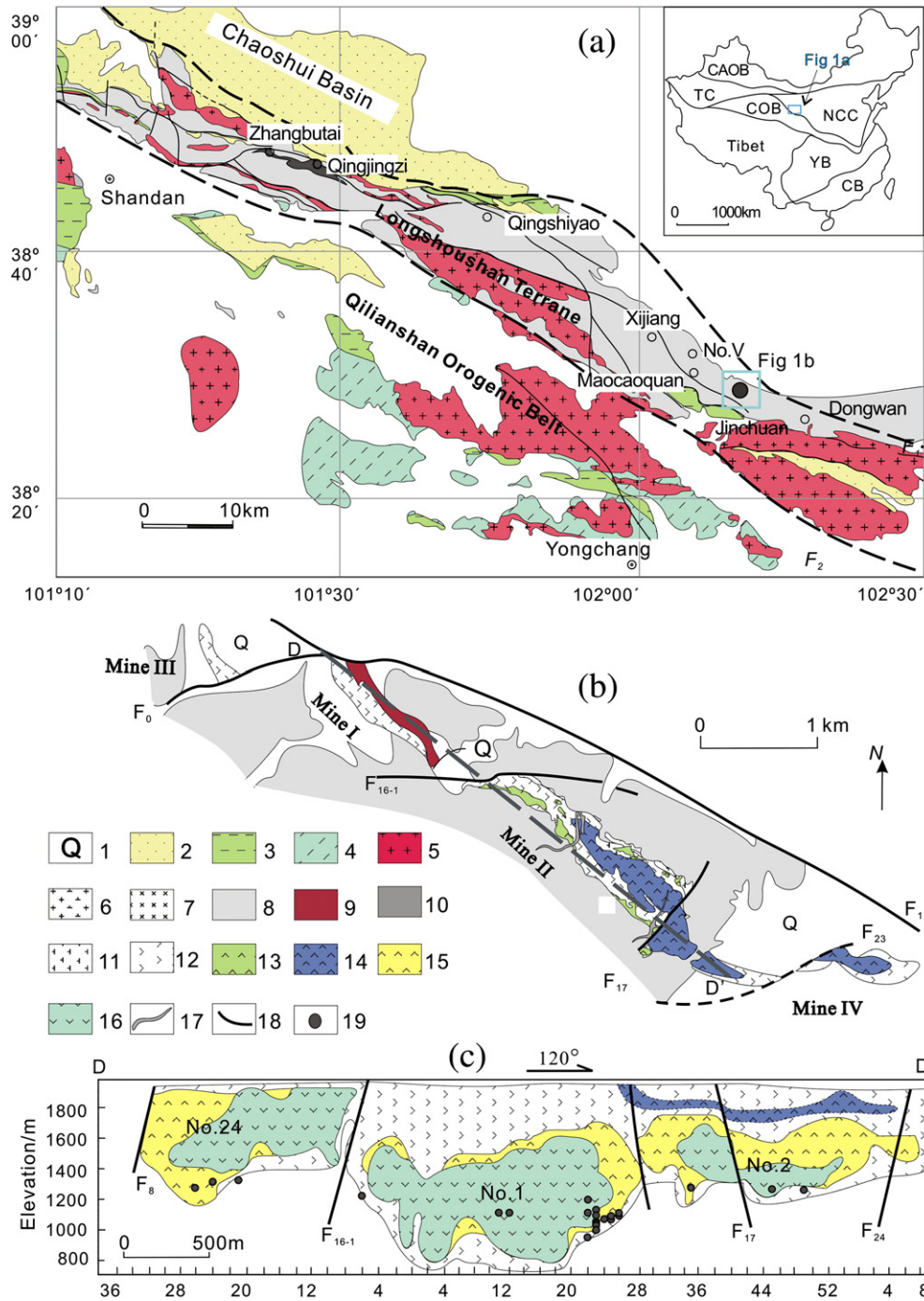


Fig. 1. Geological maps (a and b) and section (c) of the Jinchuan sulfide-bearing ultramafic intrusion (modified after Li et al., 2004). 1 – Quaternary sediments, 2 – Cretaceous sandstone, 3 – Devonian–Jurassic sedimentary rocks, 4 – Cambrian–Silurian sedimentary rocks, 5 – Paleozoic granite, 6 – Granodiorite, 7 – Migmatite, 8 – Proterozoic gneisses and meta-sedimentary rocks, 9 – Mafic-ultramafic intrusion, 10 – Oxidized No. 24 ore body, 11 – Sulfide-bearing diopside, 12 – Disseminated sulfide dunite, 13 – Net-textured sulfide-dunite, 14 – Fine-grained dunite, 15 – Medium-coarse grained lherzolite, 16 – Plagioclase lherzolite, 17 – Diabase, 18 – Faults, 19 – Sample locality. Abbreviation: NCC – Northern China Craton, CB – Cathaysia Block, TC – Tarim Craton, YB – Yangtze Block, COB – Central Orogenic Belt, CAOB – Central Asia Orogenic Belt.

3.2. Analytical methods

The concentrations of U and Th plus other trace elements in the mineral separates were determined by complete digestion of powder samples of mineral separates with HF + H₂CO₃ in Teflon capsules followed by ICP-MS analysis of the sample solutions at Chang'an University, Xi'an, China. The analytical procedures are the same as those described by Zhang et al. (2011).

Noble gases from mineral separates were extracted by vacuum stepwise heating and their concentrations and isotopic compositions were determined using an MM5400 mass spectrometer linked to the extraction system. Each sample weighed 500 mg. It was wrapped in aluminum foil and loaded into the sample chamber in a stainless steel volatile extraction system. The aluminum foil and Mo crucible were preheated at 600 °C and 1700 °C, respectively, for >24 h before they were used for sample analysis. The sample was moved to a

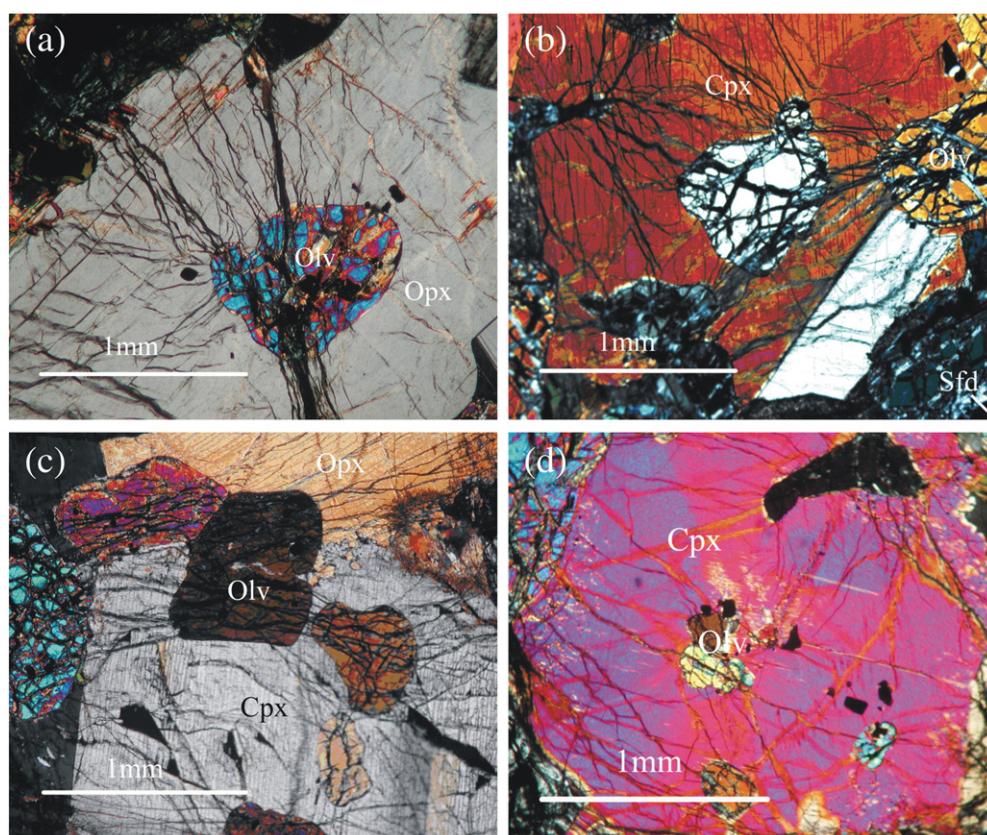


Fig. 2. Microphotographs showing the constitute minerals of Jinchuan intrusion. (a) Coarse-grained lherzolite, euhedral or round olivine, with fresh pyroxene, poikilitic olivine texture (Sample JE05). (b) Olivine is included in clinopyroxene in coarse crystal lherzolite (Sample JE09) with disseminated sulfides. (c) Plagioclase lherzolite (Sample JE16), olivine is poikilitically enclosed by pyroxene with net-textured sulfide ores, and (d) Olivine websterite (Sample JE17). Mineral abbreviations: Olv = olivine, Opx = orthopyroxene, Cpx = clinopyroxene, Sfd = sulfide.

molybdenum (Mo) crucible. The entire noble gas preparing line including the sample chamber was heated at 150 °C for more than 24 h, meanwhile the samples were preheated prior to analyses to remove loose bound (both atmospheric or possibly solar implanted) noble gases adsorbed in the sample grains.

After the extraction system had reached ultra-high vacuum (10^{-13} bars), the noble gases in sample were extracted by stepwise heating. The first fraction of volatiles released at 200 °C from the sample was discarded to get rid of possible contaminants on grain surface and exposed micro-fractures. This method can successfully eliminate noble gases of atmosphere-origin that could be absorbed on mineral surfaces and micro-fractures (Zhang et al., 2004, 2007; Ye et al., 2007; Zhang et al., 2008). The high-temperature fractions of noble gases released from the sample after a 30-minute sequential heating (200–400 °C, 400–700 °C, 700–1000 °C and 1000–1300 °C) were used for analysis. The volatiles extracted from the sample at each step were purified firstly using a spongy titanium furnace at 800 °C to remove active gases. H_2 in the volatiles was then removed by Zr–Al getters in the system at room temperature. Argon was adsorbed and isolated in a charcoal trap at liquid-nitrogen temperature (−196 °C) before He and Ne were analyzed. Other noble gases such as Ar, Kr and Xe were collected in a sintered stainless steel trap at liquid-nitrogen temperature while He and Ne were analyzed. A large portion (~60%) of the non-condensable noble gases such as He and Ne was sent to a linked MM5400 mass spectrometer. 4He was analyzed using a Faraday collector and 3He was analyzed using an electric multiple detector. Ne was analyzed next. ^{21}Ne isotope was analyzed by an electron multiplier. ^{20}Ne and ^{22}Ne were determined using a Faraday collector. Finally, the Ar was released from the trap at ice-water temperature (~0 °C), and its total volume was determined. ^{36}Ar and ^{38}Ar were analyzed using an electron multiplier. ^{40}Ar was analyzed using a Faraday cup.

The Lanzhou air standard collected from the top of Mt. Gaolan in Lanzhou city, China was used as an internal standard and its noble gas abundances and isotopes were measured before, between and after each sample. No significant fluctuation was observed in our standard during sample analysis. A hot blank was analyzed at each temperature step and the results were used for blank correction. The values of the hot blank at 1600 °C were 4He , 2.46×10^{-10} cm³.STP (in cubic centimeters at standard temperature and pressure), ^{20}Ne , 4.08×10^{-10} cm³.STP, ^{40}Ar , 1.39×10^{-8} cm³.STP. These values are similar to that of atmospheric gases. The analytical uncertainties for $^3He/^4He$, $^{20}Ne/^{22}Ne$, and $^{36}Ar/^{38}Ar$ are estimated to be within $\pm 5\%$ relative.

The effect of helium pressure on mass shift was examined and corrected. Interferences from $(^{40}Ar)^{2+}$ and $(^{12}C^{16}O_2)^{2+}$ on $(^{20}Ne)^+$ and $(^{22}Ne)^+$ were also corrected. The mass interference at $m/z = 22$ was corrected using $CO_2^+/CO_2^+ = 0.004$. Partial resolution of the $^{40}Ar^{2+}$ peak from the $^{20}Ne^+$ peak on the MM5400 mass spectrometry allows us to scan and measure $^{40}Ar^{2+}$ at $m/z = 20$. The $^{40}Ar^{2+}$ or $^{20}Ne^+$ plus $^{40}Ar^{2+}$ were determined based on estimated $^{40}Ar^{2+}/^{40}Ar^+$. The details of Ne correction are given in Ye et al. (2007). Using the above analytical procedures we (Ye et al., 2007) were able to produce similar results for the peridotites from Hawaii and lherzolite from Oki-Dogo, Japan to the values reported previously by other researchers in the world (Kurz et al., 1983; Rison and Craig, 1983; Vance et al., 1989; Nagao and Takahashi, 1993).

4. Results

The He, Ne and Ar concentrations and isotopic compositions of the Jinchuan mineral separates are listed in Table 1. The analytical uncertainties of abundances are <5% (1 σ) relative. $^3He/^4He$ ratios are

reported as the R/Ra ratios (i.e., measured $^3\text{He}/^4\text{He}$ normalized to the atmospheric ratio, $R_a = 1.39 \times 10^{-6}$).

4.1. Abundances of He, Ne and Ar

The abundances of He, Ne and Ar for each sample are the summations of all steps. ^4He , ^{20}Ne and ^{40}Ar are most dominant isotopes of the respective noble gases. They are $0.03\text{--}330.80 \times 10^{-7} \text{ cm}^3\text{.STP/g}$, $0.01\text{--}0.56 \times 10^{-7} \text{ cm}^3\text{.STP/g}$ and $0.43\text{--}1355 \times 10^{-7} \text{ cm}^3\text{.STP/g}$, respectively. ^3He and ^{36}Ar are predominantly primordial while ^{20}Ne is nonnucleogenic, they are mainly derived from the Earth's interior. The Jinchuan samples have much lower ^3He and higher ^{20}Ne and ^{36}Ar abundances (0.0017×10^{-9} , 11.75×10^{-9} and $2.98 \times 10^{-9} \text{ cm}^3\text{.STP/g}$) than that of the MORB (0.0266 , 0.17 and $0.074 \times 10^{-9} \text{ cm}^3\text{.STP/g}$) as well as the upper mantle (0.174 , 0.226 and $0.27 \times 10^{-9} \text{ cm}^3\text{.STP/g}$) (e.g., O'Nions and Tolstikhin, 1994) but rather similar to the values of continental crust and air (Allegre et al., 1987).

The He, Ne and Ar abundances of the Jinchuan samples peak at the 400–700 °C or 700–1000 °C steps (Table 1). Overall, the He, Ne and Ar abundances in the Jinchuan silicate samples are higher than the sulfide samples. The summations of average ^3He , ^{20}Ne and ^{36}Ar abundances for the Jinchuan silicate samples are $19.0 \times 10^{-13} \text{ cm}^3\text{.STP/g}$, $13.7 \times 10^{-9} \text{ cm}^3\text{.STP/g}$ and $3 \times 10^{-9} \text{ cm}^3\text{.STP/g}$, respectively. In contrast, the summations of average ^3He , ^{20}Ne and ^{36}Ar abundances for the Jinchuan sulfide samples are $8 \times 10^{-13} \text{ cm}^3\text{.STP/g}$, $4.2 \times 10^{-9} \text{ cm}^3\text{.STP/g}$ and $3.3 \times 10^{-9} \text{ cm}^3\text{.STP/g}$, respectively.

4.2. Isotopic compositions of He, Ne and Ar

The $^3\text{He}/^4\text{He}$ ratios of the Jinchuan samples vary from 0.01 to 0.83Ra with a mean of 0.27Ra. These values are similar to that of the crust and atmosphere (Allegre et al., 1987; Sarda et al., 1988) but much lower than that of typical subcontinental lithospheric mantle (SCLM) beneath North China Craton (e.g., Xu and Liu, 1997, 2002; Xu et al., 1998) where Jinchuan deposit was formed (Li and Ripley, 2011). The Jinchuan silicate separates have slightly lower $^3\text{He}/^4\text{He}$ ratios (0.01–0.57Ra, av. 0.22Ra) than coexisting sulfides (0.23–0.82Ra, av. 0.46Ra), and decrease in average ratios from olivine (av. 0.28Ra) to pyroxene (0.19Ra).

The $^{20}\text{Ne}/^{22}\text{Ne}$ ratios of the Jinchuan samples range from 8.56 to 12.22 with a mean of 10.30, which is close to the atmospheric value (9.8) (Allegre et al., 1987; Sarda et al., 1988). The $^{21}\text{Ne}/^{22}\text{Ne}$ ratios (0.026–0.043) are within the range from atmosphere (0.029) and the upper mantle (0.065) (O'Nions and Tolstikhin, 1994) to the crust (0.03–0.7). The mean $^{21}\text{Ne}/^{22}\text{Ne}$ of the Jinchuan samples (0.032) is close to the solar value (0.033) (Benkert et al., 1993).

The $^{40}\text{Ar}/^{36}\text{Ar}$ ratios vary significantly from 307 to 43285 with a mean of 9430. These values are within the range from the atmosphere (295.5) and upper mantle (4000–44000) to the crust (1650–170000) (e.g., Allegre et al., 1987; Staudacher et al., 1989; O'Nions and Tolstikhin, 1994; Burnard et al., 1997; Moreira et al., 1998; Kendrick et al., 2008). The Jinchuan silicate minerals have higher $^{40}\text{Ar}/^{36}\text{Ar}$ ratios (307–34284, av. 11633) than coexisting sulfide minerals (378–2433, av. 1108). The $^{40}\text{Ar}/^{36}\text{Ar}$ decreases in average ratio from olivine (15861) to pyroxene (9871).

He, Ne and Ar isotopic ratios of the Jinchuan samples show peak values of $^3\text{He}/^4\text{He}$, $^{21}\text{Ne}/^{22}\text{Ne}$ and $^{40}\text{Ar}/^{36}\text{Ar}$ at the 400–700 °C and 700–1000 °C steps (Table 1). The average value of $^3\text{He}/^4\text{He}$ and $^{40}\text{Ar}/^{36}\text{Ar}$ ratios at the each step decrease slightly from olivine to orthopyroxene (e.g., Samples JE11, JE16 and JE18, see details in Table 1).

4.3. The covariant of He, Ne and Ar abundance and isotopes

He, Ne and Ar released at the 200–400 °C step show different covariant in abundance and isotopes from those at the 400–700 °C,

700–1000 °C and 1000–1300 °C steps (e.g., Fig. 3a), which could be derived from different types of occurrence modes and origins (see Section 5.1 for details). Therefore, He, Ne and Ar released in the 400–700 °C, 700–1000 °C and 1000–1300 °C fractions have been summed in abundances, total isotopic compositions are recalculated, and are illustrated as a 400–1300 °C fraction in Figs. 3b, 4 and 5.

The 200–400 °C fraction of the Jinchuan samples is characterized by relatively low ^4He contents ($0.03\text{--}7.23 \times 10^{-7} \text{ cm}^3\text{.STP/g}$) as well as low and varying $^3\text{He}/^4\text{He}$ ratios (0.01–0.57Ra), and lack of correlation between $^3\text{He}/^4\text{He}$ and ^4He content (Fig. 3). The $^{20}\text{Ne}/^{22}\text{Ne}$ and $^{21}\text{Ne}/^{22}\text{Ne}$ ratios plot mainly along the mass fractionation line of air neon in the Ne isotope diagram (MFL, Sarda et al., 1988) (Fig. 4). The $^{40}\text{Ar}/^{36}\text{Ar}$ ratios of the 200–400 °C fraction vary significantly (307–40189) and show no clear correlations with relatively low ^{40}Ar contents ($0.4\text{--}39.6 \times 10^{-7} \text{ cm}^3\text{.STP/g}$, Fig. 5).

In contrast, the high-temperature fractions (400–700 °C, 700–1000 °C and 1000–1300 °C) have slightly higher $^3\text{He}/^4\text{He}$ ratios and ^4He content. These fractions collectively exhibit a negative correlation between $^3\text{He}/^4\text{He}$ and ^4He (Fig. 3), and a positive correlation between $^{40}\text{Ar}/^{36}\text{Ar}$ and ^{40}Ar content (Fig. 5). They are also characterized by relatively higher $^{20}\text{Ne}/^{22}\text{Ne}$ and $^{21}\text{Ne}/^{22}\text{Ne}$ ratios that plot more closely to the Loihi–Kilauea correlation line (L–K line, Honda et al., 1991), the MORB trend (Sarda et al., 1988; Honda et al., 1991) and the continental crust lines (CC, Allegre et al., 1987) (Fig. 4). It seems that there is to be a meaningful trend for nucleogenic He, Ne and Ar isotopes being preferentially released at high temperature steps from the silicate minerals.

5. Data interpretation and modeling

5.1. The sources and release of He, Ne and Ar

5.1.1. The sources of He, Ne and Ar

As air and cosmogenic contaminated components are likely negligible in noble gases released from the samples in the Jinchuan deposit by stepwise heating. The sulfide ore-bearing intrusions were overprinted by subsequent regional metamorphism related to subsequent tectonic events in the region (Ripley et al., 2005). Two potential sources include (1) primary noble gases from ore-forming magma, and (2) secondary noble gases, including *in-situ* radiogenic ingrowths of U, Th and K after crystallization and noble gases related to subsequent regional metamorphism or alteration.

5.1.2. The releases of He, Ne and Ar

The primary and secondary noble gases in ore formation and subsequent events may have been trapped in different sites of minerals. The primary noble gases in ore-forming magma were most likely stored in primary fluid inclusions, whereas secondary noble gases trapped in subsequent metamorphism and alteration were preserved in the secondary minerals as well as inclusions in re-crystallized minerals or healed micro-fractures, whereas radiogenic noble gases generally reside close to their production site (Matsumoto et al., 2000; Li et al., 2004; Zhang et al., 2009).

The decrepitation and degassing temperatures of different type of inclusions depend on their homogenization temperature, internal pressure of entrapment, as well as the physical strength of the host crystals (e.g., Moore et al., 2001; Villa, 2001; Zhang et al., 2004; Kendrick et al., 2006; Yang et al., 2006). The same type of inclusions in given grain size of host crystals trapped in a geological event are expected to decrepitate at temperatures similar to or higher than their incipient melting or homogenization temperatures (e.g., Moore et al., 2001; Kendrick et al., 2006; Zhang et al., 2007).

The structural defects or vacancies of mineral lattices are similar to inclusions of millimicron size, and will release their volatiles at the temperature high enough to destruct the crystal lattices, even upon melting (e.g., Matsumoto et al., 2000), therefore some of He, Ne and Ar

Table 1

Concentrations and isotopic compositions of He, Ne and Ar in silicate and sulfide minerals from the Jinchuan deposit.

No.	Rock	Mineral	Th (ppm)	U (ppm)	Temp (°C)	⁴ He (10 ^{−7} cm ³ /g)	²⁰ Ne (10 ^{−7} cm ³ /g)	⁴⁰ Ar (10 ^{−7} cm ³ /g)	³ He/ ⁴ He (Ra)	²⁰ Ne/ ²² Ne	²¹ Ne/ ²² Ne	³⁸ Ar/ ³⁶ Ar	⁴⁰ Ar/ ³⁶ Ar
<i>Mine I</i>													
JL012	Pyt	Opx	0.266	0.079	200–400	1.970 ± 0.130	0.0154 ± 0.0044	1.54 ± 0.11	0.075 ± 0.007	10.54 ± 0.31	0.0321 ± 0.0076	0.2120 ± 0.0180	955.0 ± 78.1
					400–700	14.610 ± 0.980	0.1580 ± 0.0180	11.80 ± 0.86	0.493 ± 0.005	11.71 ± 0.08	0.0308 ± 0.0024	0.2820 ± 0.0280	2698.0 ± 331.1
					700–1000	5.210 ± 0.350	0.0386 ± 0.0061	81.00 ± 5.50	0.123 ± 0.004	9.65 ± 0.09	0.0409 ± 0.0021	0.2250 ± 0.0120	13054.0 ± 605.7
					1000–1300	6.780 ± 0.460	0.0449 ± 0.0080	11.58 ± 0.78	0.134 ± 0.005	9.61 ± 0.10	0.0293 ± 0.0039	0.2190 ± 0.0140	4638.0 ± 245.4
JL036	MO	Sfd			200–400	3.160 ± 0.210	0.0221 ± 0.0040	46.60 ± 3.10	0.704 ± 0.004	9.39 ± 0.31	0.0334 ± 0.0012	0.1720 ± 0.0029	1258.0 ± 31.3
					400–700	17.700 ± 1.200	0.0271 ± 0.0045	55.70 ± 3.70	0.853 ± 0.008	9.69 ± 0.23	0.0347 ± 0.0009	0.2045 ± 0.0077	2170.0 ± 60.3
					700–1000	1.720 ± 0.120	0.0458 ± 0.0046	161.00 ± 11.00	0.543 ± 0.010	9.25 ± 0.12	0.0354 ± 0.0008	0.2330 ± 0.0140	3847.0 ± 171.6
					1000–1300	0.005 ± 0.001	–	7.54 ± 0.51	–	–	–	0.1891 ± 0.0060	306.0 ± 6.4
JS046	NTO	Sfd			200–400	1.281 ± 0.088	0.0354 ± 0.0061	0.84 ± 0.07	0.600 ± 0.019	10.56 ± 0.11	0.0284 ± 0.0002	0.2710 ± 0.0270	767.0 ± 91.0
					400–700	4.570 ± 0.310	0.0140 ± 0.0049	0.44 ± 0.03	0.527 ± 0.011	9.39 ± 0.57	0.0315 ± 0.0020	–	655.0 ± 92.6
					700–1000	3.970 ± 0.270	0.0090 ± 0.0039	2.20 ± 0.15	0.379 ± 0.007	7.77 ± 0.25	0.0302 ± 0.0009	0.2070 ± 0.0160	2329.0 ± 129.5
JL056	MO	Sfd			200–400	11.950 ± 0.800	0.0203 ± 0.0042	18.00 ± 1.20	0.389 ± 0.004	9.82 ± 0.43	0.0333 ± 0.0021	0.1652 ± 0.0062	573.0 ± 22.6
					400–700	8.120 ± 0.540	0.0126 ± 0.0038	8.63 ± 0.58	0.461 ± 0.007	9.66 ± 0.45	0.0352 ± 0.0009	0.1756 ± 0.0041	417.0 ± 11.5
					700–1000	23.800 ± 1.600	0.0160 ± 0.0088	22.30 ± 1.50	0.268 ± 0.005	7.85 ± 0.68	0.0410 ± 0.0032	0.1728 ± 0.0017	1840.0 ± 42.7
<i>Mine II</i>													
JE012	d-Lhz	Opx			200–400	1.630 ± 0.260	0.0477 ± 0.0072	4.75 ± 0.33	0.109 ± 0.003	10.87 ± 0.10	0.0306 ± 0.0003	0.2220 ± 0.0098	1779.0 ± 75.6
					400–700	12.390 ± 0.830	0.0849 ± 0.0120	53.10 ± 3.50	0.272 ± 0.007	11.28 ± 0.07	0.0326 ± 0.0004	0.2620 ± 0.0130	8627.0 ± 477.9
					700–1000	23.200 ± 1.500	0.0256 ± 0.0059	94.20 ± 6.30	0.347 ± 0.009	9.54 ± 0.21	0.0346 ± 0.0010	0.2099 ± 0.0074	25437.0 ± 1010.0
					1000–1300	5.150 ± 0.350	0.0124 ± 0.0047	91.00 ± 6.10	0.153 ± 0.011	7.81 ± 0.40	0.0328 ± 0.0015	0.2150 ± 0.0087	31055.0 ± 956.5
JE021	d-Lhz	Cpx	0.096	0.023	200–400	0.067 ± 0.012	0.1020 ± 0.0150	20.50 ± 1.50	0.314 ± 0.026	10.62 ± 0.10	0.0278 ± 0.0004	0.1973 ± 0.0014	337.0 ± 3.0
					400–700	5.730 ± 0.390	0.5070 ± 0.0520	105.40 ± 8.60	0.232 ± 0.011	10.70 ± 0.05	0.0272 ± 0.0002	0.2105 ± 0.0087	471.0 ± 21.4
					700–1300	2.800 ± 0.200	0.0510 ± 0.0130	20.80 ± 1.40	0.123 ± 0.008	10.29 ± 0.18	0.0301 ± 0.0008	0.1970 ± 0.0054	478.0 ± 15.8
JE022	d-Lhz	Opx	0.165	0.038	200–400	2.320 ± 0.160	0.0495 ± 0.0088	11.99 ± 0.82	0.107 ± 0.009	11.27 ± 0.22	0.0293 ± 0.0010	0.2050 ± 0.0100	649.0 ± 42.2
					400–700	14.090 ± 0.950	0.1510 ± 0.0230	135.70 ± 9.40	0.287 ± 0.005	10.51 ± 0.09	0.0267 ± 0.0025	0.1980 ± 0.0012	7591.0 ± 351.5
					700–1000	18.000 ± 1.200	0.0520 ± 0.0110	132.90 ± 8.90	0.297 ± 0.004	9.01 ± 0.22	0.0265 ± 0.0008	0.1821 ± 0.0053	22151.0 ± 1090.0
					1000–1300	3.130 ± 0.210	0.0317 ± 0.0077	81.60 ± 5.50	0.161 ± 0.006	9.84 ± 0.28	0.0294 ± 0.0009	0.1758 ± 0.0050	5803.0 ± 265.2
JE052	c-Lhz	Opx			200–700	25.600 ± 1.700	0.1870 ± 0.0400	217.00 ± 15.00	0.279 ± 0.006	8.85 ± 0.27	0.0286 ± 0.0009	0.2046 ± 0.0019	14732.0 ± 178.3
					700–1100	158.000 ± 11.000	0.2420 ± 0.0720	488.00 ± 33.00	0.199 ± 0.009	12.11 ± 0.26	0.0368 ± 0.0010	0.2064 ± 0.0054	29637.0 ± 912.8
					1100–1300	15.300 ± 1.000	0.1390 ± 0.0160	884.00 ± 59.00	0.165 ± 0.008	8.91 ± 0.04	0.0311 ± 0.0019	0.2162 ± 0.0048	36184.0 ± 759.9
JE082	c-Lhz	Opx			200–400	3.170 ± 0.210	0.0268 ± 0.0071	32.20 ± 2.20	0.131 ± 0.010	8.75 ± 0.22	0.0259 ± 0.0009	0.1810 ± 0.0170	16071.0 ± 1535.0
					400–700	108.300 ± 7.200	0.0770 ± 0.0270	512.00 ± 34.00	0.226 ± 0.003	9.41 ± 0.17	0.0412 ± 0.0007	0.2199 ± 0.0072	25459.0 ± 873.2
					700–1000	135.400 ± 9.100	0.0890 ± 0.0440	344.00 ± 23.00	0.179 ± 0.005	10.23 ± 0.35	0.0466 ± 0.0014	0.2342 ± 0.0085	35092.0 ± 1369.0
					1000–1300	87.100 ± 5.800	0.0550 ± 0.0320	275.00 ± 18.00	0.134 ± 0.006	10.99 ± 0.62	0.0238 ± 0.0013	0.2149 ± 0.0054	36170.0 ± 1241.0
JE092	d-Lhz	Opx			200–400	0.534 ± 0.038	0.0392 ± 0.0054	10.40 ± 0.71	0.176 ± 0.011	11.47 ± 0.18	0.0339 ± 0.0006	0.1988 ± 0.0061	362.0 ± 6.7
					400–700	10.990 ± 0.740	0.0640 ± 0.0100	20.70 ± 1.40	0.197 ± 0.004	11.56 ± 0.11	0.0346 ± 0.0032	0.2094 ± 0.0055	1305.0 ± 29.8
					700–1000	111.900 ± 7.500	0.0830 ± 0.0430	137.00 ± 9.20	0.171 ± 0.003	10.92 ± 0.45	0.0460 ± 0.0021	0.2068 ± 0.006	11393.0 ± 283.7
					1000–1300	25.800 ± 1.700	0.0260 ± 0.0130	35.80 ± 2.40	0.144 ± 0.004	8.67 ± 0.31	0.0342 ± 0.0011	0.1985 ± 0.0063	4024.0 ± 92.2
JE111	d-Lhz	Olv	0.09	0.018	200–400	1.033 ± 0.072	0.0295 ± 0.0047	10.96 ± 0.74	0.155 ± 0.008	10.88 ± 0.17	0.0328 ± 0.0005	0.1998 ± 0.0059	431.0 ± 11.3
					400–700	21.900 ± 1.500	0.0001 ± 0.0001	162.00 ± 11.00	0.282 ± 0.006	9.77 ± 0.29	0.0347 ± 0.0010	0.2061 ± 0.0065	7172.0 ± 165.0
					700–1000	51.000 ± 3.400	0.0238 ± 0.0091	69.40 ± 4.60	0.339 ± 0.004	9.22 ± 0.52	0.0381 ± 0.0019	0.1937 ± 0.0063	6632.0 ± 173.1
					1000–1300	9.930 ± 0.670	0.0588 ± 0.0092	237.00 ± 16.00	0.153 ± 0.003	9.12 ± 0.10	0.0342 ± 0.0008	0.1958 ± 0.0067	9265.0 ± 198.3

JE112	d-Lhz	Opx			200–400	5.640 ± 0.380	0.0783 ± 0.0081	6.18 ± 0.44	0.145 ± 0.004	10.83 ± 0.10	0.0323 ± 0.0004	0.2140 ± 0.0120	2332.0 ± 24.9
					400–700	8.630 ± 0.580	0.0252 ± 0.0062	12.39 ± 0.85	0.097 ± 0.004	10.90 ± 0.25	0.0360 ± 0.0009	0.2560 ± 0.0210	7222.0 ± 579.2
					700–1000	137.100 ± 9.200	0.0420 ± 0.0240	220.00 ± 15.00	0.189 ± 0.005	10.59 ± 0.86	0.0561 ± 0.0045	0.2305 ± 0.0061	32868.0 ± 1242.0
					1000–1300	40.700 ± 2.700	0.0460 ± 0.0130	155.00 ± 10.00	0.139 ± 0.004	9.57 ± 0.35	0.0363 ± 0.0012	0.2108 ± 0.0076	20331.0 ± 813.2
JE132	d-Lhz	Opx			200–400	0.664 ± 0.047	0.0640 ± 0.0100	6.38 ± 0.45	0.240 ± 0.011	12.22 ± 0.12	0.0343 ± 0.0001	0.2058 ± 0.0064	371.0 ± 9.9
					400–700	80.100 ± 5.400	0.1400 ± 0.0420	103.70 ± 7.10	0.225 ± 0.006	11.22 ± 0.44	0.0405 ± 0.0012	0.2422 ± 0.0087	5681.0 ± 200.0
					700–1000	152.000 ± 10.000	0.0740 ± 0.0320	144.40 ± 9.60	0.217 ± 0.004	10.17 ± 0.41	0.0455 ± 0.0013	0.1986 ± 0.0065	13835.0 ± 319.6
					1000–1300	84.900 ± 5.700	0.0750 ± 0.0310	157.00 ± 10.00	0.164 ± 0.003	10.23 ± 0.30	0.0450 ± 0.0015	0.212 ± 0.0067	13886.0 ± 299.9
JE161	Lhz	Olv	0.108	1.148	200–400	1.920 ± 0.130	0.0096 ± 0.0034	9.82 ± 0.66	0.250 ± 0.008	9.15 ± 0.45	0.0291 ± 0.0009	0.1929 ± 0.0094	5672.0 ± 167.9
					400–700	54.700 ± 3.700	0.1550 ± 0.0240	540.00 ± 37.00	0.393 ± 0.007	9.58 ± 0.04	0.0327 ± 0.0002	0.2470 ± 0.0089	34666.0 ± 1397.0
					700–1000	93.100 ± 6.300	0.1670 ± 0.0720	372.00 ± 25.00	0.379 ± 0.009	8.89 ± 0.48	0.0318 ± 0.0018	0.2260 ± 0.0230	54154.0 ± 834.0
					1000–1300	12.810 ± 0.860	0.0338 ± 0.0057	384.00 ± 26.00	0.202 ± 0.005	8.90 ± 0.15	0.0338 ± 0.0006	0.2300 ± 0.0140	42362.0 ± 2715.0
JE162	Lhz	Opx	0.258	0.472	200–400	4.740 ± 0.320	0.0359 ± 0.0059	11.65 ± 0.82	0.182 ± 0.012	10.59 ± 0.25	0.0302 ± 0.0007	0.2804 ± 0.0054	5613.0 ± 104.4
					400–700	112.100 ± 7.500	0.0610 ± 0.0280	478.00 ± 32.00	0.159 ± 0.002	11.47 ± 0.47	0.0638 ± 0.0024	0.2220 ± 0.0120	40973.0 ± 2000.0
					700–1000	77.200 ± 5.200	0.0400 ± 0.0230	261.00 ± 17.00	0.252 ± 0.002	8.33 ± 0.50	0.0405 ± 0.0022	0.2219 ± 0.0083	49012.0 ± 2166.0
					1000–1300	25.000 ± 1.700	0.0178 ± 0.0083	82.10 ± 5.50	0.138 ± 0.004	9.90 ± 0.67	0.0453 ± 0.0032	0.2030 ± 0.0110	20810.0 ± 1140.0
JE172	Pyt	Opx			200–400	0.990 ± 0.069	0.1700 ± 0.0180	0.43 ± 0.03	0.044 ± 0.002	11.47 ± 0.05	0.0295 ± 0.0002	0.3650 ± 0.0250	394.0 ± 30.3
					400–700	26.800 ± 1.800	0.0660 ± 0.0140	4.15 ± 0.30	0.321 ± 0.007	11.48 ± 0.18	0.0329 ± 0.0004	0.2010 ± 0.0280	753.0 ± 41.6
					700–1000	2.820 ± 0.190	0.0305 ± 0.0055	19.70 ± 1.30	0.042 ± 0.003	10.29 ± 0.16	0.0399 ± 0.0004	0.1995 ± 0.0065	2780.0 ± 115.7
					1000–1300	1.011 ± 0.070	0.0065 ± 0.0029	3.46 ± 0.23	0.053 ± 0.003	10.92 ± 0.75	0.0334 ± 0.0030	0.1982 ± 0.0074	1342.0 ± 49.9
JE181	c-Lhz	Olv			200–400	7.230 ± 0.480	0.0089 ± 0.0360	39.60 ± 2.60	0.230 ± 0.010	10.54 ± 0.39	0.0398 ± 0.0017	–	40189.0 ± 6736.0
					400–700	56.500 ± 3.800	0.0099 ± 0.0061	308.00 ± 21.00	0.362 ± 0.004	9.50 ± 1.20	0.0515 ± 0.0067	0.2450 ± 0.0150	40793.0 ± 436.5
					700–1000	24.000 ± 1.600	0.0430 ± 0.0110	162.00 ± 11.00	0.335 ± 0.008	10.32 ± 0.43	0.0311 ± 0.0013	0.2900 ± 0.1500	58139.0 ± 25727.0
					1000–1300	9.050 ± 0.610	0.0780 ± 0.0110	324.00 ± 22.00	0.251 ± 0.004	8.92 ± 0.08	0.0229 ± 0.0004	0.2110 ± 0.0110	38458.0 ± 2511.0
JE182	c-Lhz	Cpx			200–400	0.067 ± 0.007	0.0071 ± 0.0029	1.85 ± 0.13	0.012 ± 0.002	9.37 ± 0.57	0.0281 ± 0.0018	0.2130 ± 0.0140	3079.0 ± 152.7
					400–700	35.200 ± 2.400	0.0620 ± 0.0270	421.00 ± 28.00	0.274 ± 0.007	8.87 ± 0.39	0.0327 ± 0.0017	0.2280 ± 0.0120	25455.0 ± 1400.0
					700–1000	24.800 ± 1.700	0.0330 ± 0.0110	107.10 ± 7.20	0.226 ± 0.008	9.38 ± 0.42	0.0345 ± 0.0014	0.2192 ± 0.0084	28324.0 ± 1076.0
					1000–1300	11.310 ± 0.760	0.0178 ± 0.0051	87.10 ± 5.80	0.194 ± 0.006	8.98 ± 0.33	0.0336 ± 0.0011	0.2290 ± 0.0110	34900.0 ± 2010.0
JE183	c-Lhz	Cpx			200–400	0.804 ± 0.056	0.0305 ± 0.0051	8.70 ± 0.59	0.136 ± 0.007	11.17 ± 0.26	0.0271 ± 0.0006	0.2160 ± 0.0100	3953.0 ± 144.7
					400–700	25.400 ± 1.700	0.0520 ± 0.0180	35.00 ± 2.400	0.198 ± 0.005	10.13 ± 0.26	0.0331 ± 0.0008	0.2130 ± 0.0180	16915.0 ± 1812.0
					700–1000	30.200 ± 2.000	0.0420 ± 0.0100	169.00 ± 17.00	0.166 ± 0.015	10.20 ± 0.32	0.0290 ± 0.0009	–	–
					1000–1300	13.480 ± 0.980	0.0173 ± 0.0040	64.60 ± 4.30	0.166 ± 0.005	9.37 ± 0.26	0.0281 ± 0.0007	0.2800 ± 0.1000	34676.0 ± 12754.0
JE191	Lhz	Olv			200–400	0.233 ± 0.019	0.4000 ± 0.0350	36.60 ± 2.50	0.340 ± 0.024	11.68 ± 0.04	0.0281 ± 0.0001	0.1980 ± 0.0170	470.0 ± 37.7
					400–700	0.682 ± 0.048	0.0677 ± 0.0070	1.81 ± 0.13	0.312 ± 0.017	11.28 ± 0.17	0.0274 ± 0.0004	0.3200 ± 0.0550	784.0 ± 137.7
					700–1000	5.150 ± 0.350	0.0128 ± 0.0049	22.50 ± 1.50	0.269 ± 0.006	9.01 ± 0.34	0.0302 ± 0.0011	0.2340 ± 0.0160	12707.0 ± 870.5
					1000–1300	2.040 ± 0.140	0.0142 ± 0.0050	15.00 ± 1.00	0.187 ± 0.007	9.13 ± 0.42	0.0284 ± 0.0008	0.2220 ± 0.0110	14502.0 ± 584.4
JE192	Lhz	Opx			200–400	0.030 ± 0.005	0.0210 ± 0.0042	5.48 ± 0.37	0.570 ± 0.150	9.85 ± 0.28	0.0280 ± 0.0008	0.1953 ± 0.0061	307.0 ± 11.5
					400–700	20.500 ± 1.400	0.0410 ± 0.0110	58.20 ± 3.90	0.156 ± 0.006	10.17 ± 0.37	0.0319 ± 0.0010	0.1944 ± 0.0050	2062.0 ± 55.5
					700–1000	24.400 ± 1.600	0.0340 ± 0.0110	78.80 ± 5.30	0.163 ± 0.004	8.85 ± 0.32	0.0299 ± 0.0007	0.1927 ± 0.0056	4546.0 ± 142.8
					1000–1300	16.400 ± 1.100	–	34.70 ± 2.30	0.144 ± 0.005	–	–	–	–
JE206	MO	Sfd			200–400	3.860 ± 0.260	0.0562 ± 0.0070	11.22 ± 0.79	0.233 ± 0.004	11.21 ± 0.18	0.0327 ± 0.0007	0.2099 ± 0.0072	403.0 ± 8.9
					400–700	5.540 ± 0.370	0.0193 ± 0.0490	8.00 ± 0.53	0.301 ± 0.005	9.85 ± 0.24	0.0314 ± 0.0008	0.1946 ± 0.0063	367.0 ± 8.0
					700–1000	4.500 ± 0.300	0.0081 ± 0.0310	3.89 ± 0.26	0.257 ± 0.007	10.05 ± 0.46	0.0357 ± 0.0012	0.1973 ± 0.0057	468.0 ± 11.1
					200–400	28.200 ± 1.900	0.0155 ± 0.0078	11.45 ± 0.77	0.349 ± 0.011	10.52 ± 0.71	0.0410 ± 0.0023	0.1524 ± 0.0067	1772.0 ± 63.4
JE146	MO	Sfd			400–700	20.100 ± 1.300	0.0058 ± 0.0028	4.66 ± 0.31	0.432 ± 0.005	10.75 ± 0.46	0.0477 ± 0.0015	0.1970 ± 0.0110	1028.0 ± 46.9
					700–1000	21.700 ± 1.500	0.0100 ± 0.0058	4.07 ± 0.28	0.409 ± 0.014	11.77 ± 0.95	0.0643 ± 0.0037	0.2140 ± 0.0100	1778.0 ± 49.6

System blanks were 9×10^{-9} cm³.STP for ⁴He, $0.6\text{--}8 \times 10^{-12}$ cm³.STP for ²⁰Ne, and $2\text{--}6 \times 10^{-12}$ cm³.STP for ³⁶Ar for the different extraction temperatures. The isotopic ratios of blanks are similar to atmospheric ones. Errors are 1 σ . Rock types: Lhz = lherzolite; c-Lhz = coarse-grained lherzolite; d-Lhz = disseminated sulfide-bearing lherzolite; Pyt = pyroxenite. Ore types: NTO = net-textured ore, MO = massive ore. Minerals: Olv = olivine, Opx = orthopyroxene, Cpx = clinopyroxene, Sfd = sulfide.

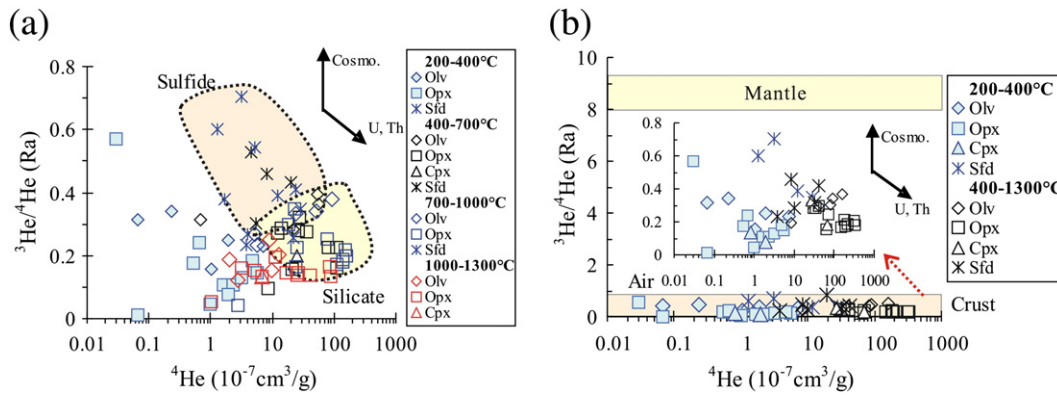


Fig. 3. Plots of $^3\text{He}/^4\text{He}$ vs. ^4He contents ($10^{-7} \text{ cm}^3/\text{g}$) at each steps (a) and summing as two fractions (b). Arrows indicate changes in isotope ratios caused by the addition of cosmogenic (Cosmo.) and radiogenic (U, Th) components (Ikeda et al., 2001); other abbreviations are the same as in Fig. 2.

released at the 1000–1300 °C fraction are likely derived from the crystal structural defects or vacancies. Olivine and pyroxene from ultramafic rocks are brashy and weak in physical strength relative to quartz and are thus more easily be cracked with fluid inclusions more easily decrepitated and releasing their volatiles (Zhang et al., 2009).

Accordingly, noble gases released at the 200–400 °C fraction can be interpreted to be derived from secondary gas–liquid inclusions entrapped during subsequent metasomatic or alteration processes. The primary plus radiogenic noble gases are mainly collected in the fractions of 400–700 °C, 700–1000 °C and 1000–1300 °C (Yang et al., 2006), they were summed and presented as a “400–1300 °C” fraction in the Figs. 3b, 4 and 5.

5.2. The noble gases in the secondary processes

The noble gases in secondary processes include noble gases trapped in metamorphism or alteration, as well as radiogenic products.

5.2.1. The noble gases in post-magmatic metamorphism or alteration

The noble gases in metamorphism or secondary alteration had been trapped in secondary fluid–gas inclusions. These types of fluid inclusions were homogenized to a liquid between 235 and 365 °C or decrepitated above ~230 °C (e.g., Yang et al., 2006). Significantly varying $^3\text{He}/^4\text{He}$ (0.01–0.57Ra) and $^{40}\text{Ar}/^{36}\text{Ar}$ (307–40189), and lack of correlations with low ^4He ($0.03\text{--}7.23 \times 10^{-7} \text{ cm}^3/\text{STP/g}$) and

^{40}Ar contents ($0.4\text{--}39.6 \times 10^{-7} \text{ cm}^3/\text{STP/g}$) in the 200–400 °C fraction (Figs. 3, 5, and 6) are consistent with the interpretation that they are mainly derived from secondary fluid inclusions with negligible radiogenic ^4He and ^{40}Ar , which could be overprinted by a secondary air saturated fluid (ASF) during a hydrothermal metasomatism or alteration event (Fig. 6).

The higher $^{20}\text{Ne}/^{22}\text{Ne}$ and lower $^{21}\text{Ne}/^{22}\text{Ne}$ ratios in the 200–400 °C fraction fall mainly between the MFL and L–K line (Fig. 4). The ^{20}Ne enriched components can be attributed to either exotic primordial noble gases in the Earth (possibly representing solar neon), or mass fractionation of neon from an original mantle or atmosphere neon (Fig. 4) (e.g. Sarda et al., 1988). Negligible radiogenic ^4He , ^{21}Ne and ^{40}Ar suggest that atmospheric fluids have been involved for a short period, i.e., a secondary alteration could be responsible for noble gas isotopic signatures, because regional metamorphism by the collision event of Qilian block and the North China Craton (Song et al., 2006b) could form the accumulation of radiogenic ^4He and ^{40}Ar for ~450 Ma. This is supported by the presence of two fluids involved in lower temperature hydrothermal alteration, as constrained by oxygen and hydrogen isotopes (e.g., Ripley et al., 2005).

5.2.2. The radiogenic noble gases

^4He , ^{40}Ar , ^{21}Ne and ^{22}Ne can be produced by radiogenic or nucleogenic reactions, and usually remain close to their production sites (e.g., Staudacher et al., 1986). The net results of radiogenic in-growths in igneous rocks will result in a negative correlation between $^3\text{He}/^4\text{He}$ and ^4He content, positive correlations between ^{21}Ne and ^4He content (e.g., Sarda et al., 1988; Honda et al., 1991; Shaw et al., 2001) and between $^{40}\text{Ar}/^{36}\text{Ar}$ and ^{40}Ar content, with a trend towards lower $^{20}\text{Ne}/^{22}\text{Ne}$ and higher $^{21}\text{Ne}/^{22}\text{Ne}$ ratios (e.g., Kennedy et al., 1990; Ballentine and Burnard, 2002). These features are all present in the

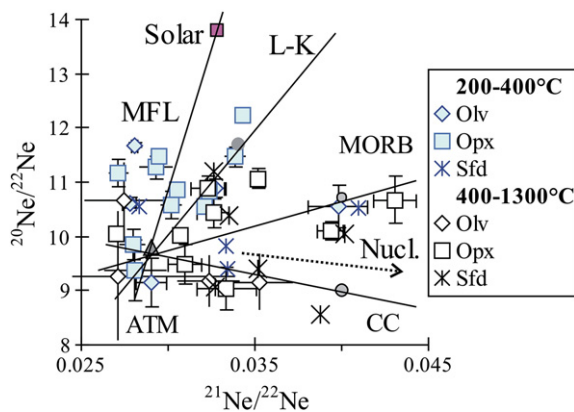


Fig. 4. Ne three-isotope diagram ($^{20}\text{Ne}/^{22}\text{Ne}$ vs. $^{21}\text{Ne}/^{22}\text{Ne}$). The linear mass fractionation line of air neon is shown as MFL (Sarda et al., 1988). Air: atmosphere (Allegre et al., 1987; Sarda et al., 1988). Solar: solar wind (Benkert et al., 1993). CC: continental crust line (Allegre et al., 1987). MORB: mid-ocean ridge basalts (Sarda et al., 1988). L–K: Loihi–Kilauea trend (Honda et al., 1991; Hiyagon et al., 1992). Nucl.: nucleogenic component (Kennedy et al., 1990; Ballentine and Burnard, 2002). Other abbreviations are the same as in Fig. 2.

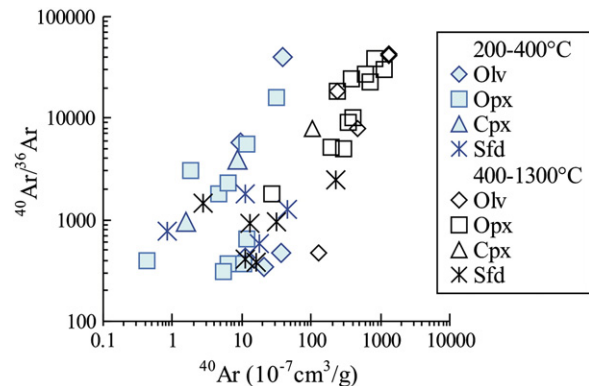


Fig. 5. Plot of $^{40}\text{Ar}/^{36}\text{Ar}$ vs. ^{40}Ar contents ($10^{-7} \text{ cm}^3/\text{g}$). Abbreviations are the same as in Fig. 2.

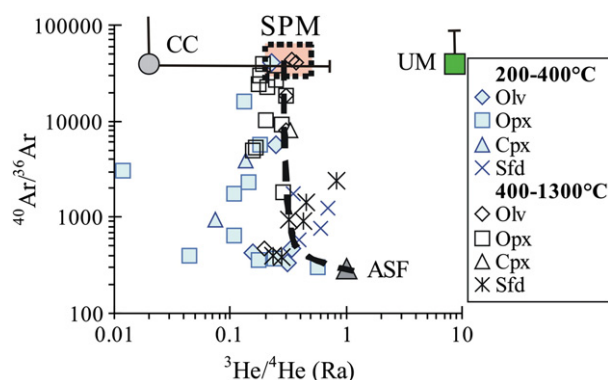


Fig. 6. Plot of $^3\text{He}/^4\text{He}$ vs. $^{40}\text{Ar}/^{36}\text{Ar}$. UM – upper mantle, from O’Nions and Tolstikhin (1994) and Staudacher et al. (1989), ASF – air saturated fluid, from Ozima and Podosek (2002); CC: continental crust (Allegre et al., 1987); SPM – the source of primary magma. Other abbreviations are the same as in Fig. 2.

400–1300 °C fraction of noble gases released from the Jinchuan samples (Figs. 3–5), suggesting that radiogenic ingrowth of $^4\text{He}^*$, $^{21}\text{Ne}^*$ and $^{40}\text{Ar}^*$ (where an asterisk denotes radiogenic) are important in the Jinchuan samples.

The amounts of radiogenic $^4\text{He}^*$ and $^{40}\text{Ar}^*$ ingrowth in the Jinchuan samples can be calculated using the average concentrations of U and Th in mineral separates determined by us and whole rock samples (Chai and Naldrett, 1992a; Song et al., 2006a) and the crystallization age of the Jinchuan intrusion (831 Ma) (Zhang et al., 2010). Calculated contents of $^{40}\text{Ar}^*$ in both silicate and sulfide using the U and Th contents of whole rocks are av. 10.4×10^{-7} and 6.6×10^{-7} cm³.STP/g, respectively, which are lower than measured contents of ^{40}Ar (av. 263.9×10^{-7} and 40.21×10^{-7} cm³.STP/g); $^{40}\text{Ar}/^{36}\text{Ar}$ ratios range from 469.8 to 42962 when radiogenic $^{40}\text{Ar}^*$ have been corrected. However calculated $^4\text{He}^*$ contents in both silicate (av. 197×10^{-7} cm³.STP/g) and sulfide (av. 161×10^{-7} cm³.STP/g) using the U and Th contents of whole rocks are higher than measured contents of ^4He (av. 60.9×10^{-7} and 13.2×10^{-7} cm³.STP/g), which imply (1) He diffusion, or (2) higher contents of U and Th in whole rocks than in the mineral separates.

The mineral separates from the Jinchuan deposit show lower U (av. 0.396 and 0.196 ppm in Olv and Pyx, respectively) and Th (0.098 and 0.229) contents than the whole rocks (Chai and Naldrett, 1992a; Song et al., 2006a), indicating that U and Th could be mainly concentrated in minor interstitial phases. Calculated $^4\text{He}^*$ contents using the U and Th contents of minerals are av. 5.3×10^{-7} cm³.STP/g in Olv and av. 3.1×10^{-7} cm³.STP/g in Pyx, $^3\text{He}/^4\text{He}$ ratios range from 1.6 to 5.1 when radiogenic $^4\text{He}^*$ has been corrected. Slightly higher $^3\text{He}/^4\text{He}$ values (av. 0.46Ra) of sulfide minerals than silicate minerals (0.22Ra) are caused by lower U and Th contents in sulfide minerals than silicate minerals (Song et al., 2009, 2012; Li and Ripley, 2011). So, both He diffusion and lower U and Th contents in mineral separates could be responsible for above variation of noble gas isotopes in the Jinchuan samples.

5.3. The primary noble gases in the ore-forming magma

The ore-forming magma of the Jinchuan deposit is interpreted to have formed by a high-MgO (up to 18.5 wt.%) tholeiitic basaltic magma which were originated from a SCLM (Chai and Naldrett, 1992a, 1992b; de Waal et al., 2004). It started to crystallize in a staging chamber (Li et al., 2004; Song et al., 2006b, 2009). Multiple emplacements of basaltic magma charged with olivine are important in the formation of the Jinchuan ore-bearing intrusion (Song et al., 2006b, 2009, 2012; Tang et al., 2009). The sulfide liquids of different compositions, represented by different ore zones, were possibly formed by multiple stages of sulfide segregation from a basaltic magma in a staging chamber (Li et al., 2004; Song et al., 2006b, 2009).

The primary noble gases in the ore-forming magma were intrinsically stored in primary fluid inclusions and crystal structural defects and vacancies of magmatic mineral lattices during crystallization of the silicate and sulfide minerals. Primary fluid inclusions, including crystallized polyphase inclusions and glass-sulfide inclusions in Jinchuan deposit (Yang et al., 2006), could be dominant occurrence modes of primary noble gases, and are easy to keep closed in olivine and pyroxene, because olivine and pyroxene are the important retentive minerals of noble gases, and because diffusivity or exchange between inclusions with olivine and pyroxene minerals is likely insignificant due to silicate glasses or amorphous carboniferous film precipitates lining walls of inclusions in magmatic minerals (e.g., Burnard et al., 1997; Kamenetsky et al., 2002).

The primary noble gases could be represented by the 400–1300 °C fraction based on the released temperature range similar to the incipient melting or homogenization temperatures of primary fluid inclusions (Yang et al., 2006). The noble gases in the 400–1300 °C fraction are characterized by the highest ratios of $^3\text{He}/^4\text{He}$, $^{21}\text{Ne}/^{22}\text{Ne}$ and $^{40}\text{Ar}/^{36}\text{Ar}$ (Figs. 3, 4 and 5), and are most close to the isotopic compositions of the primary noble gases involved in magma evolution and ore formation at Jinchuan. Low $^3\text{He}/^4\text{He}$ ratios and slightly negative correlation with ^4He contents (Fig. 3), high $^{40}\text{Ar}/^{36}\text{Ar}$ ratios and positive correlation with ^{40}Ar contents (Fig. 5) indicate that the presence of radiogenic $^4\text{He}^*$ and $^{40}\text{Ar}^*$. $^{20}\text{Ne}/^{22}\text{Ne}$ and $^{21}\text{Ne}/^{22}\text{Ne}$ ratios fall mainly in the range between upper mantle (MORB line) and continental crust (CC line, Fig. 4) (e.g., Allegre et al., 1987; Sarda et al., 1988). The variations of He and Ar isotopes depict a ternary mixing among the mantle or SCLM, continental crust (CC) and ASF (Fig. 6). These variations may be caused by (1) source characteristics of the SCLM where the Jinchuan magma formed, plus radiogenic accumulation after crystallization, (2) magma degassing, and (3) contamination or mixing with crustal and atmospheric components.

5.3.1. Source characteristics and radiogenic ingrowths

The SCLM is a unique noble gas reservoir due to a long term isolation from the mantle convection, in which diverse noble gas elemental and isotopic components are distinct from those in the sublithospheric reservoirs and present a more radiogenic $^3\text{He}/^4\text{He}$ of 8.39×10^{-6} (~6.1 Ra) (e.g., Dunai and Baur, 1995; Gautheron and Moreira, 2002). The Jinchuan ultramafic intrusion was formed beneath the SCLM of the Northern China Craton (Li and Ripley, 2011), where $^3\text{He}/^4\text{He}$ ratios range from 0.15 to 15.78 Ra with a mean of 4.8 Ra (Xu and Liu, 1997; Xu et al., 1998), which is lower than the values of global values of SCLM (6.1 Ra) (Dunai and Baur, 1995; Gautheron and Moreira, 2002). The highest ratio of $^3\text{He}/^4\text{He}$ (5.1 Ra) in Jinchuan silicate samples after radiogenic $^4\text{He}^*$ has been corrected is slightly higher than mean (4.8 Ra) in the SCLM of the Northern China Craton.

He in mantle-derived rocks must be restricted to primary He (i.e., primordial He escaping from the lower mantle and crustal He) plus *in situ* products of radiogenic He from U and Th-series isotopes. He isotopes can be used to trace either the magma source of intrusive rocks or post-intrusive radioactive decay of uranium and thorium. The presence of a negative correlation between $^3\text{He}/^4\text{He}$ and ^4He in the 400–1300 °C fraction (Fig. 3) indicates that radiogenic $^4\text{He}^*$ could be responsible for lower $^3\text{He}/^4\text{He}$ ratios in the Jinchuan deposit. The simulation results of radiogenic $^4\text{He}^*$ addition from U and Th decay are illustrated in Fig. 7. In our calculations we used the average $^3\text{He}/^4\text{He}$ value (8.39×10^{-6}) of a global SCLM to represent the initial ratio. Clearly, the low $^3\text{He}/^4\text{He}$ values can be explained by significant radiogenic ingrowth of $^4\text{He}^*$ from average U and Th contents of lherzolite, pyroxenite and all types of rocks in the Jinchuan deposit (Song et al., 2006b). Therefore, both low $^3\text{He}/^4\text{He}$ ratios in the ore-forming magma source from the underlying SCLM and ingrowth of radiogenic $^4\text{He}^*$ could be responsible for lower $^3\text{He}/^4\text{He}$ ratios observed in the Jinchuan samples. Our interpretation is supported by

available Nd–Os isotopes for the Jinchuan intrusion (see data compiled by Li and Ripley, 2011).

5.3.2. The magma degassing

The magma degassing is a common process in volatile-saturated magma. The residual volatiles in the magma chamber increases in $^4\text{He}/^{40}\text{Ar}_{\text{Air}}$ ratios (as a degassing proxy) with ^4He contents decreasing during basaltic magma degassing, because helium is more soluble than argon in basaltic melts (Sarda and Graham, 1990; Yamamoto and Burnard, 2005), where the $^{40}\text{Ar}_{\text{Air}}$ is the measured ^{40}Ar “corrected” for atmospheric contamination, i.e.,

$$^{40}\text{Ar}_{\text{Air}} = \left(^{40}\text{Ar}/^{36}\text{Ar} - ^{40}\text{Ar}/^{36}\text{Ar}_{\text{Air}} \right) \times ^{36}\text{Ar}$$

The plot of $^4\text{He}/^{40}\text{Ar}_{\text{Air}}$ and ^4He (Fig. 8) suggests that the effect of magma degassing could be ignored in the 400–1300 °C fraction (e.g., O’Nions and Tolstikhin, 1994; Trierloff et al., 2000), as $^4\text{He}/^{40}\text{Ar}$ variation could be caused by radiogenic contributions.

On the other hand, lighter noble gases diffuse rapidly out into the magma during generation and migration of magma in the mantle. He has higher diffusivity than Ne and Ar, which can result in decoupling of He with Ne and Ar in natural samples (e.g., Jambon et al., 1986; Carroll and Stolper, 1993; Burnard, 2004; Yamamoto et al., 2009).

5.3.3. The contributions of crustal and atmospheric origins

The He, Ne and Ar isotopes depict a ternary mixture among the SCLM, CC and ASF, the source of primary magma (SPM) was a mixture of SCLM and CC components, and then added by ASF (Fig. 6). The proportions of each component can be estimated by He and Ar isotopic mixing models.

He has very low concentrations in the atmosphere and the shortest atmospheric residence time of 1 to 10 million years, and can escape from any system at elevated temperatures (e.g., Lupton, 1983; Marty et al., 1989; Stuart et al., 1994), hence, atmospheric He recycled into the mantle is negligible (e.g., Porcelli and Wasserburg, 1995), thus, SCLM and crust are considered as the only two end-members of the He isotopic mixing model to calculate the proportions of SCLM (X_M) and crust components (X_{cc}) in SPM according to the conventional formula:

$$X_M = \left[\left(^3\text{He}/^4\text{He} \right) - \left(^3\text{He}/^4\text{He} \right)_{\text{CC}} \right] / \left[\left(^3\text{He}/^4\text{He} \right)_{\text{SCLM}} - \left(^3\text{He}/^4\text{He} \right)_{\text{CC}} \right]$$

Continental crust (CC) has originally $^3\text{He}/^4\text{He}$ values of 2×10^{-8} (0.02Ra, e.g., Ballentine and Burnard, 2002), while SCLM has $^3\text{He}/^4\text{He}$ values of 8.39×10^{-6} (~6.1 Ra) (e.g., Dunai and Baur, 1995; Gautheron and Moreira, 2002). The calculated results of the proportions of

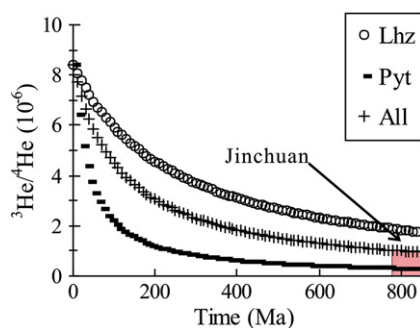


Fig. 7. Calculated $^3\text{He}/^4\text{He}$ ratios decrease by radiogenic $^4\text{He}^*$ ingrowths according to U and Th contents of whole rocks of lherzolite (Lhz), pyroxenite (Pyt) and all type of rocks (All).

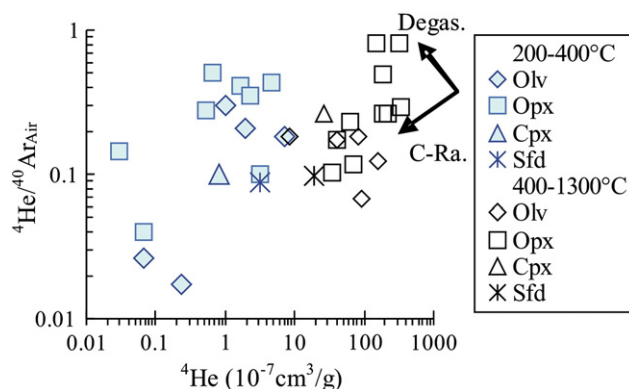


Fig. 8. Plot of $^4\text{He}/^{40}\text{Ar}_{\text{Air}}$ vs. ^4He contents ($10^{-7} \text{ cm}^3/\text{g}$). Degas. — magma degassing, C-Ra. — contributions of radiogenic. Other abbreviations are the same as in Fig. 2.

SCLM-derived He (X_M) in the Jinchuan samples by the He isotopic mixing model indicate that SCLM-derived He constitutes 27%–85% with a mean of 46% in the 400–1300 °C fraction, after radiogenic $^4\text{He}^*$ has been corrected from silicate mineral separates. About 54% continental crust fluid was added to the Jinchuan ore-forming magmatic system.

Air saturated fluid (ASF) shows air-like noble gas isotopes. Ar has accumulated much more in the atmosphere than He and Ne over Earth's history, and is a powerful geochemical tracer for atmospheric origin components (Fisher, 1997). On the other hand, continental crust shows a $^{40}\text{Ar}/^{36}\text{Ar}$ ratio (28000) (e.g., Allegre et al., 1987) similar to that of the upper mantle (20600) (e.g., Staudacher et al., 1989; O’Nions and Tolstikhin, 1994), hence, the proportion of ASF addition has been estimated based on an Ar isotopic mixing model of continental crust and air.

The calculation results of Ar isotopic mixing model show that ASF in the Jinchuan samples is about 74% in the 200–400 °C fractions, and 63% in the 400–1300 °C fraction. Combined with the results of He isotope mixing model, the average proportion of ASF, CC and SCLM mixed in the primary noble gases (400–1300 °C fraction) of the Jinchuan samples are estimated as 63%, 20% and 17%, respectively.

5.4. The noble gas isotopic constrain on ore-genesis

Holistic lower $^3\text{He}/^4\text{He}$ ratios in silicate and sulfide minerals in the Jinchuan deposit plus radiogenic simulation calculations suggest that ore-forming primary magma might be derived from SCLM, and a proportion of continental crustal fluid could have been added into primary magma in deep staging magma chamber. In addition, no obvious evidences of crustal origin in terms of major elements, trace elements and sulfur isotopes were found (e.g., Ripley et al., 2005), and our calculated proportion (20%) of crustal component is higher than results (mostly 5–15%) suggested by Sr–Nd isotopes (Li and Ripley, 2011). We proposed that continental crustal fluids instead of whole rocks were added to magma during magma generation, emplacement and subsequent crystallization.

The assimilation of the crustal fluids and ASF could also increase the oxygen fugacity, decrease the solubility of S in magma (Wallace and Carmichael, 1992; Mavrogenes and O’Neill, 1999), and thus trigger sulfide saturation and segregation.

Atmospheric component could be added into mantle-derived magma by subduction, deeply circulated metasomatic ASF or secondary alteration (e.g., Fisher, 1997; Bach and Niedermann, 1998). Therefore, ASF related metasomatism (especially for the 400–1300 °C fraction) could be responsible for the atmospheric component. Postmagmatic hydrothermal alteration may have been triggered by heat from large Paleozoic granite plutons formed during the collision between the North Qilian suture zone and the Langshoushan terrane at ~450 Ma (Song et al., 2006a), and has affected the ultramafic rocks

and associated sulfide ores of the Jinchuan intrusion (Chai and Naldrett, 1992a; Barnes and Tang, 1999; de Waal et al., 2004; Li et al., 2004; Ripley et al., 2005; Lehmann et al., 2007), then were followed by subsequent regional metamorphism (Song et al., 2006b, 2009; Tang et al., 2009) as well as hydrothermal alteration (Ripley et al., 2005), which are partly responsible to atmospheric component in 200–400 °C fraction.

6. Conclusions

Important conclusions from this study are summarized below.

- (1) The noble gas isotopic compositions of volatiles released from the Jinchuan samples at relatively low-T (200–400 °C) record low-T postmagmatic hydrothermal overprinting in the intrusion.
- (2) The isotopic compositions of noble gases extracted from the Jinchuan samples at higher temperatures in the 400–1300 °C fraction are consistent with significant radiogenic ingrowths after crystallization of the intrusion from magma derived from subcontinental lithospheric mantle.
- (3) Mass balance calculations of He and Ar isotopic mixing models indicate that a crustal fluid of about 20% were added to magma during magma generation, emplacement and subsequent crystallization; it could affect the oxygen fugacity and the solubility of S in magma, and trigger sulfide liquid segregation. About 63% ASF were added into the Jinchuan parental magma during contact metamorphism suggested previously by some researchers.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.chemgeo.2012.07.023>.

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